Advanced Radio License Notes

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This are the notes from following the advanced course on HamRadio.au.

2 Electronics and Circuit Theory

2.1 Impedance

2.1.1 Introduction

Impedance is how a circuit stores or resists AC (or any changing) energy. If the impedances within a circuit aren't matched, then power gets reflected back instead of going where it is needed.

Impedance is the combination of resistance and reactance. Resistance is the restriction of the "flow" of energy, and converts that restricted energy to heat. This is measured in Ohms Ω. Reactance can be thought of as a flywheel, it stores energy temporarily and releases it. This is what capacitors and inductors do. This is also measured in Ohms. This means that impedance is the total opposition to current.

> **Why this matters.** The transmitter expects to see 50Ω of impedance. The coax is also 50Ω. If the antenna isn't also close to 50Ω, then the power from the transmitter gets reflected, which is what SWR (standard wave ratio) measures.

2.1.2 Reactance

There are two types of reactance. One is found in inductors, and the other in capacitors.

Inductors (any coil of wire) opposes changes in current, i.e. their reactance increases with frequency

$$X_L = 2\pi f L.$$

This means that the higher the frequency, the more and inductor blocks the signal. This is why they are used in low-pass filters, they all for low frequencies but block high ones.

Capacitors opposes changes in voltage. Their reactance decreases with frequency

$$X_C = \frac{1}{2\pi f C}.$$

The higher the frequency, the easier it passes through the capacitors. This is why a capacitor in series blocks DC but passes RF.

2.1.3 Impedance formula

We can then write impedance as

$$Z = R + jX,$$

where R is the resistance, X is the impedance, and j is used in place of i for imaginary numbers due to i being used for current. The resistance part is where the energy is consumed, and the reactive part is where the energy is stored and returned. The total impedance is then given by the magnitude of this complex number

$$|Z| = \sqrt{R^2 + X^2}$$

> **Example.** An antenna has $R = 50\Omega$ and $X = 25\Omega$. The total impedance magnitude is
>
> $$\begin{aligned} Z &= \sqrt{2500 + 625} \\ &= \sqrt{3125} \\ &\approx 56\Omega \end{aligned}$$

2.2 Resonance

2.2.1 Introduction

When an inductor and capacitor are combined, at one specific frequency their reactances become equal and therefore cancel each other out. This is called resonance, and when this happens the circuit behaves as purely resistive. This is how a single frequency is selected while rejecting everything else.

> **Radio Example.** Tuned circuits within a receiver use this resonance to select a station to hear, and reject everything else. Antennas are resonant at a specific frequency, which is where it works best.

The formula to find the resonant frequency is given by

$$f_r = \frac{1}{2\pi\sqrt{LC}}.$$

This says that for a given inductor L and capacitor C, there's exactly one frequency where they resonate. This is an important formula.

2.2.2 Series and Parallel Resonance

Series and parallel arrangements of L and C result in opposite behaviour at resonance.

Table 1: Behaviour of circuits at resonance

	Series LC	Parallel LC
Impedance	Minimum (just R)	Maximum
Current	Maximum	Minimum from source
Used for	Passing a desired frequency (band-pass)	Blocking a desired frequency (or selecting it) (band-stop)

> **Radio Example.** A series-resonant circuit in an antenna trap passed the trap's frequency through, while a parallel-resonant circuit in a receiver's IF (intermediate frequency) stage has high-impedance at the desired frequency, which develops a large signal voltage across it.

2.2.3 Q-Factor

The Q-factor described how selective a resonant circuit is. A higher A means a very sharp tuning (with narrow bandwidth), and low Q means a very broad tuning (wide bandwidth)

$$Q = \frac{f_r}{\text{Bandwidth}}$$

For a series RLC, $Q = \frac{X_L}{R}$, so less resistance means higher Q.

> **Bandwidth Rule.** Bandwidth is $\frac{f_r}{Q}$, so a circuit with $Q = 100$ at 7 MHz has a bandwidth of 70 kHz, which is enough to cover a portion of the 40m band. A crystal filter width $Q = 50000$ at 9 MHz has a bandwidth of only 180 Hz, which is perfect for CW.

2.3 Filters

2.3.1 Introduction

Filters are extremely important in radio. They are what selects the frequencies that get let through and what get blocked. Without filters, a receiver would hear literally everything at the same time, and the transmitter would spray harmonics all over the spectrum (which happens to be illegal).

2.3.2 Types of Filter

There are four basic types of filter.

Table 2: Basic types of filter

Filter Type	Behaviour	Where it is in radios
Low-pass	Passes low frequencies, blocks high frequencies	After the transmitter PA (power amplifier). It blocks harmonics before reaching the antenna
High-pass	Passes high frequencies, block low frequencies	At a TV antenna input. It blocks HF signals, but lets VHF/UHF TV through
Band-pass	Passes a specific range, blocks everything else	IF (intermediate frequency) filters in a receiver. It selects the signal bandwidth that you want.
Band-stop (notch)	Blocks a specific range, passes everything else	Notch filter to remove an interfering carrier.

2.3.3 LC Filter

Filters use inductors and capacitors in combinations to make a specific filter. From above, we know

- Inductors block high frequencies (reactance increases with frequency)
- Capacitors block low frequencies (reactance decreases with frequency)

> **Low-pass Filter.** A low-pass filter wants to block high frequencies, and allow low frequencies through. This means that the circuit would put an inductor in series (to block high frequencies), and capacitors to ground (short high frequencies to ground).

2.3.4 Filter Steepness

In practice, filters can't instantly go from passing a frequency to blocking another. The change is gradual, and this is affected by the number of LC sections (called a pole). Each pole adds 20 dB/decade of rolloff. The higher this number, the more steep the dropoff, as it is saying that per decade (1 decade is a ratio of 10 between two numbers on a logarithmic scale), the scale drops by 20 dB.

> **Example.** Consider a transmitter that produces a 7 MHz signal. The 2nd harmonic at 14 MHz and 3rd harmonic at 21 MHz must be supressed. A good 5-element low-pass filter with a cutoff just about 7 MHz can suppress the 2nd harmonic by 40+ dB and the 3rd by 60+ dB. This is the difference between a clean signal and an interference complaint.

A typical transmitter low-pass filter has 5-poles (100 dB/decade, a steep slope). For critical applications, a 7-pole filter is used (140 dB/decade, a very steep slope).

2.3.5 Crystal Filters

Crystal filters have very high Q factors (10 000 to 100 000+). Multiple crystals combined in a so-called ladder filter give the sharp selectivity for IF stages

- 2.4 kHz bandwidth - for SSB reception
- 500 Hz bandwidth - for CW reception
- 6 kHz bandwidth - for AM reception

In modern radios however, crystals are being replaced with DSP (digital signal processing) filters, as they allow changing the bandwidth on the fly.

2.3.6 Filter Response Shapes

Different mathematical designs give different trade-offs. Butterworth designs have a smooth, flat passband with no ripple. This is used where consistent gain across the passband matters. Chebyshev has a steeper rolloff than Butterworth, but some some ripple in the passband. This is used where sharp selectivity matters more than a perfectly flat response.

2.4 Amplifier Classes

2.4.1 Introduction

The final stage of a transmitter is a power amplifier. This amplifier will have some bias, and depending on how that bias is configured changes two important things. First is the linearity - how faithfully does the amplifier reproduce the signal shape (critical for SSB and AM). The second is efficiency - how much of the DC power becomes RF (all excess is heat).

2.4.2 Classes

These are the different classes of amplifiers

> **Important.** Class C amplifiers are not suitable for SSB or AM. Both of these carry information in the variation in amplitude, so when class C clips the signal information is destroyed. It is fine for FM and CW as these modes have a constant amplitude.

Most HF transceivers use Class AB for the final PA. It is linear enough to handle SSB without excessive distortion, and efficient enough that a 400W output doesn't require a large heatsink. If a Class AB amp is overdriven (i.e. mic gain is too high), it starts producing intermodulation distortion. This is a "splatter" that interferes with adjacent frequencies.

Table 3: Amplifier classes

Class	How it works	Efficiency	Linearity	Used for
A	Transistor conducts all the time (360°). Always on	25 – 50%	Excellent	Low-power stages, driver stages
AB	Conducts most of the time (180°-360°). Slightly biased below full-on	50 – 65%	Very good	Standard choice for SSB power amplifiers
B	Conducts exactly half the time (180°). Needs push-pull pairs	65 – 78%	OK	Push-pull audio amps
C	Conducts less than half the time (<180°). Very ”punchy”	75 – 85%	Poor	FM and CW only

2.4.3 Decibels

Decibels allows for working with ratios by simply adding and subtracting instead of multiplying and dividing.

Table 4: Shortcuts for working with decibels

dB	Power ratio	Voltage ratio	Notes
+3 dB	×2	×1.41	Double the power
+6 dB	×4	×2	Double the voltage
+10 dB	×10	×3.16	Ten times the power
+20 dB	×100	×10	Ten times the voltage
-3 dB	×0.5	×0.71	Half the power

Example. An amplifier has 13 dB of gain. That is $10 + 3 = \times 10 \times \times 2 = \times 20$ power. Put in 20W, get out 400W.

Info. The factor is 10 for power, 20 for voltage. That is because power is proportional to the voltage squared.
Power formula: $G_{\mathrm{dB}} = 10 \log_{10}(P_{\mathrm{out}}/P_{\mathrm{in}})$.
Voltage formula: $G_{\mathrm{dB}} = 20 \log_{10}(P_{\mathrm{out}}/P_{\mathrm{in}})$.

2.5 Oscillators

2.5.1 Introduction

An oscillator generates a continuous AC signal at a specific frequency. This is what the transmitter uses to generate the carrier signal, and the receiver needs one to convert signals to the IF. An oscillator is an amplifier where some of the output is fed back into the input in-phase (positive feedback).

In order for this to work correctly, two criteria must be met (known as the Barkhasuen criteria). The first is that the loop gain must be greater than or equal to 1. The fed-back signal must be at least as strong as what is needed at the input. The second criteria is that the fed-back signal must be in phase so that it can reinforce the original signal.

2.5.2 Common Types

Colpitts Oscillator: uses two capacitors and one inductor. The two capacitors form a voltage divider for the feedback. It is very popular in VFOs (Variable Frequency Oscillators) as it is stable and easy to tune by varying one of the capacitors.

$$f = \frac{1}{2\pi\sqrt{L \cdot \frac{C_1 C_2}{C_1 + C_2}}}$$

Hartley Oscillator: uses one capacitor and a tapped inductor (or two inductors). The tapped inductor provides the feedback. This is less common in modern equipment.

Crystal Oscillator: uses a quartz crystal as the frequency determining element. The crystal has an extremely high Q (10 000+), making it incredibly stable. It is used for reference oscillators in synthesisers, BFO (Beat Frequency Oscillator) in CW/SSB receivers, and any application where frequency accuracy is critical.

> **Radio Connection.** When tuning a HF radio, usually you are not changing the oscillator directly, rather modern radios use a PLL (Phase-Locked Loop) synthesiser. This is a crystal oscillator that provides the stable reference, and the PLL multiplies it to the frequency that you want.

2.5.3 PLL Frequency Synthesiser

The PLL is how modern radios generate precise, tunable frequencies. In simple terms:

1. A crystal reference provides an accurate, stable signal (e.g. 10 kHz)
2. A VCO (Voltage-Controlled Oscillator) generates the output frequency
3. A divider divides the VCO output down
4. A phase detector compares the divided output with the reference
5. If they don't match, it adjusts the VCO until they do.

$$f_{\text{out}} = N \times f_{\text{ref}}$$

> **Example.** Consider a reference $f_{\text{ref}} = 10$ kHz and $N = 1450$. The output is then $f_{\text{out}} = 14.500$ MHz (20 metres). Changing N to 1451 gives $f_{\text{ref}} = 14.510$ MHz. This is how a tuning knob works in 10 kHz steps.

2.5.4 Mixers

A mixer takes two signals and produces their sum and difference frequencies

$$f_{\text{out}} = f_1 + f_2 \quad \text{and} \quad f_{\text{out}} = f_1 - f_2.$$

This is how receivers convert the signal you want to the IF, and how transmitters shift the generated signal to the operating frequency. The best type of mixer is the double-balanced mixer. This makes filtering much easier as it suppresses both input signals at the output, passing main the sum and difference products.

2.6 AGC and Power Supplies

2.6.1 Introduction

AGC (automatic gain control) automatically adjusts the gain of a receiver so that the audio level remains roughly constant. It works by

1. The receiver measures the strength of the incoming signal
2. This creates a control voltage
3. The control voltage adjusts the of the RF and IF amplifier stages

This allows for less gain with strong signals, and more gain with weaker signals. AGC can be either fast or slow. Fast AGC has a quick attack and release, which is best for CW and digital modes, and allows for quick recovery between elements. Slow AGC still has quick attack, however the release is slowed. This is best for SSB voice, and doesn't "pump" between syllables. Pumping is volume surging between words.

2.6.2 Power Supplies for Radio

Most amateur HF transceivers need a regulated 13.8V DC supply. There are two main types of power supply.

Linear power supply: simple and electrically quiet. Works by dropping excess voltage across a regulator, which turns into heat. The efficient is dependent on the input and output voltages, i.e. efficiency = $V_{\text{out}}/V_{\text{in}}$. The result of all this is that it is preferred for HF because of low noise, however requires a large mains transformer.

Switch-mode power supply (SMPS): must more efficient and lighter, but can generate RF interference. It works by rapidly switching DC on and off (50-500 kHz), which is then usually passed through a filter to reduce noise. This is significantly cheaper and lighter than linear power supplied.

> **EMC Consideration.** If a receiver has buzzing or whining that changes with tuning, the SMPS might be the cause. Ferrite chokes on the DC power leads dampen this, and keep the power leads away from the antenna feedline.

3 Transmitters and Receivers

3.1 The Superheterodyne Receiver

Trying to make a very narrow filter that can be tuned across the entire HF spectrum (3-30 MHz) would result in having tto retune the filter each time the frequency is change. Superheterodyne (superhet) receivers solve this problem by instead of adjusting the filter, it modifies the signal such that it falls within the filter.

A superhet receiver is broken down into stages

1. RF Preselector: a broad filter that roughly selects the band of interest, and rejects signals that are obviously out of band.
2. Mixer + Local Oscillator (LO): The mixer combines the signal with the LO. This transforms the signal to match the IF. For example, if you are tuned to 14.200 MHz and the IF is 9 MHz, then the LO runs at 23.200 MHz (23.200-14.200=9.000 MHz)
3. IF Filter: this is fixed at 9 MHz (or whatever IF), and doesn't change when you tune. It provides the sharp selectivity required. Crystal or DSP filters are used here to determine the bandwidth.
4. IF Amplifier: This boosts the filtered signal (controlled by AGC).
5. Detector: Converts the IF signal back to audio. For SSB, this is a product detector (mixer with a BFO).
6. Audio Amplifier: Drives the output audio device.

3.1.1 The Image Problem

Because the mixer doesn't know which signal produced the 9 MHz difference, there is a second ”image” of the signal placed at 2 times the IF. For example, if the LO is at 23.2 MHzm both a signal at 14.2 MHZ and a signal at 32.2 MHz produce a 9 MHz signal.

$$f_{\text{image}} = f_{\text{signal}} \pm 2 \times f_{\text{IF}}.$$

The RF preselector must be good enough to reject the image. A higher IF makes this easier as the separation between the frequencies is higher. High-end receivers combat this by first converting to a high IF (i.e. 45-70 MHz) to create the separation between the signal and the image, then converts to the final IF (9 MHz or 455 MHz), which allows for narrow and affordable crystal filters.

3.2 SSB

3.2.1 Introduction

Regular AM transmits a carrier plus two sidebands, but since the carrier carries no information and the sidebands carry the same information this is wasteful. Instead, SSB (single sideband modulation) removes the carrier and one sideband, which saves power and bandwidth. Generally, the process is

1. Audio amplifier: the amplified output of the microphone
2. Balanced modulator: mixes the audio with a carrier frequency. It still generates both sidebands, but suppresses the carrier. This is now called DSB-SC (Double Sideband, Suppressed Carrier)
3. Crystal filter: a very sharp band-pass filter that passes one of the sidebands and rejects the other
4. Mixer: shift the SSB signal from the filter up to the operating frequency
5. Power amplifier (Class AB): boosts the signal to the desired output power
6. Low-pass filter: removes the harmonics before transmitting

> **Convention.** The global convention or which sideband is used is dependant on the frequency. When it is less than 10 MHz, the LSB (lower sideband) is used, and for 10 MHz and above, USB (upper sideband) is used.

3.2.2 The Phasing Method

Instead of using an expensive crystal oscillator, this method uses phase shifting to cancel one sideband mathematically. It requires precise 90°shifts in both the audio and RF paths. This is popular in SDR (software defined radio) as phase shifting is easy to do in software.

3.2.3 Balanced Modulator

The balanced modulator is a mixer that is specifically designed so that the carrier leaks through as little as possible. The output contains the two sidebands, but almost no carrier. That is why it is called balanced.

3.3 FM

3.3.1 Introduction

Whilst SSB (and AM) modulates the amplitude of the signal to carry voice, FM varies the frequency, and the amplitude remains constant. FM is used for most VHF/UHF communication (2m and 70cm repeaters), as it is simpler, more noise resistant, and sounds better than SSB for local communications.

For FM, there are three important values. The first is deviation (δf), and is a measure of how far the frequency swings from the carrier. For amateur narrow-band FM, this is ± 5 kHz. The next value is modulation index β, and this is the deviation divided by the audio frequency $\beta = \delta f / f_m$. The final value is the bandwidth (Carson's rule), and is given by

$$\text{BW} \approx^2 (\delta f + f_m).$$

> **Example.** With a ± 5 kHz deviation and 3 kHz maximum audio, bandwidth is given by $2(5+3) = 16$ kHz. This is why FM channels on 2m are spaced 12.5 or 25 kHz apart.

One of the biggest advantages of FM is that when two stations transmit on the same frequency, the stronger signal wins completely. The weaker signal is suppressed, not mixed in with the strong one.

Info. The capture effect is why FM works so well for repeaters. If two station accidentally transmit at the same time, you hear the stronger one clearly and not a mix of both. With SSB, you'd hear both simultaneously.

3.3.2 Pre- and De-emphasis

FM noise gets worse at higher audio frequencies. To prevent this, transmitters (pre-emphasis) boost high audio frequencies before transmission, and receivers (de-emphasis) cuts high audio frequencies after demodulation. The boost and cut cancel out for the unwanted signal, but the noise (which was not boosted) gets cut. This results in better signal-to-noise.

3.4 Receiver Performance

3.4.1 Introduction

On a busy HF band, a receiver might be dealing with signals ranging from barely detectable (-130 dBm), to extremely strong (-10 dBm), which is a range of over 100 million to one. Good receivers are able to handle both simultaneously.

3.4.2 Sensitivity

Sensitivity is limited by noise. Every receiver generates some internal noise, and the signal must be stronger than this noise floor to be heard. The noise figure of the front is the key specification, as it tells you how much noise the receiver adds compared to the perfect receiver.

Friis Formula. The noise figure of the first stage dominates, because noise from later stages gets divided by the gain of earlier stages. This is why a low-noise preamp at the antenna makes a big difference.

$$NF_{\text{total}} \approx NF_1 + \frac{NF_2 - 1}{G_1} + \frac{NF_3 - 1}{G_1 G_2} + \cdots .$$

3.4.3 Intermodal Distortion (IMD)

When two strong signals hit a receiver, any non-linearity creates phantom signals at new frequencies. The most troublesome are third-order products

$$f_{\text{IMD}} = 2f_1 - f_2 \quad \text{and} \quad 2f_2 - f_1.$$

This falls close to the original signals and cannot be filtered out.

Example. Two strong stations on 14.200 and 14.210 MHz create a phantom signal at $2(14.200) - 14.210 - 14.190$ MHz. If you are trying to hear a weaker station at the frequency, you would only be able to hear the IMD product.

IP3 (third-order intercept point) is the key specification from strong signal handling. Higher IP3 is better. This is the theoretical point where the IMD products would equal the wanted signal. This is never reached in practice.

3.4.4 Reciprocal Mixing

In the real world, no oscillator is perfect, they all have some sort of phase noise (random frequency jitter). When a strong off-channel signal mixes with this phase noise, it creates at the IF that buries weak signals. This means that a better oscillator gives better reception, and why more expensive radios perform better on crowded bands.

3.4.5 Dynamic Range

Dynamic range is the difference between the weakest signal the receiver can detect and the strongest signal it can handle without distortion. More dynamic range means better performance on busy bands. A receiver with 90 dB of dynamic range can simultaneously handle signals differing by 90 dB, which is about 1 billion to one in power.

4 Propagation

4.1 The Ionosphere

4.1.1 Introduction

The ionosphere is a layer of charged gas in the upper atmosphere (60-500 km), which is caused by ultraviolet radiation from Sol stripping away electrons. Because of this charge, HF signals are able to bounce off of this layer and travel thousands of kilometres.

4.1.2 The Layers

There are three main layers in the ionosphere.

The D Layer (60-90km) absorbs signals, but only exists during the daylight. It mostly affects signals below 10 MHz, but reflects barely any signals.

The E Layer (90-150km) is able to reflect signals up to 10 MHz (roughly) during the day, but also disappears during the night.

The F Layer (150-500km) is the most important layer for HF communication. During the day it is separated into a F1 and F2 layer, however recombines during the night to a single F layer. The F2 layer is the highest and has the most ionisation, and hence capable of the longest skip distances.

> **The day/night cycle of HF.** During the day, the D layer absorbs low-band signals (80m, 40m), but the F2 layer supports higher bands (20m, 15m, 10m). At night, the D layer vanishes opening up the low bands, while the F layer weakens, which closes the high bands. This affects what bands to use at what times.

4.2 Selecting the Right Frequency

4.2.1 Introduction

For any given path between two stations, there's a range of frequencies that will work. Too high of a frequency and the signal will punch through the ionosphere, too low and the D layer absorbs it.

4.2.2 Maximum Usable Frequency (MUF)

The MUF is the highest frequency the ionosphere will return to Earth for a specific path. Above this, signal passes straight through into space. This frequency varies with time of day, season, solar cycle, and path length.

4.2.3 Minimum Usable Frequency (LUF)

This is the lowest frequency that provides a usable signal. Below this, the D layer absorption is too strong. The LUF is higher during the day and lower during the night.

4.2.4 Frequency of Optimum Traffic (FOT)

This is the most optimum frequency, given by

$$\text{FOT} = 0.85 \times \text{MUF}$$

This is the frequency for the most reliable communication.

Table 5: The best bands for different times

Time	Best Bands	Notes
Daytime	20m, 17m, 15m, 12m, 10m	High MUF, D-Layer absorbs low bands
Evening	40m, 30m, 20m	D Layer fading, F layer still good
Night	160m, 80m, 40m	No D layer absorption, low MUF
Sunrise/sunset	160m, 80m (grey line)	Special propagation, see below.

4.2.5 Skip Distance and Skip Zone

When a signal goes up to the ionosphere and comes back down, it obviously does not come back down directly next to the transmitter. We call the minimum distance where the skywave signal returns to Earth the skip distance, and the area between where the ground wave fades out and where the skywave returns the skip zone (also known as dead zone). In general, a higher frequency corresponds to a longer skip distance. On 10m, the skip might be 1000+ km, where as 40m it might be 300km.

4.3 Special Propagation

4.3.1 Sporadic E

Sometimes a random patches of ionisation in the E layer can reflect signals well into the VHF range, especially on 6 metres (50 MHz), and even 2m. This is most common in the summer, but is very unpredictable. Single hop distances can be 500-2300 km.

4.3.2 Grey-Line Propagation

The grey-line is the sunset/sunrise boundary on Earth. Along this line, the D layer is either yet to form (sunrise), or has just disappeared (sunset), but the F layer still has enough ionisation to reflect signals. This creates a low-absorption ”duct” that reflects signals on the low bands. This is a brief window, and both the transmitting and receiving stations must be on or near the grey line.

4.3.3 Tropospheric Ducting

Not all propagation has to involve the ionosphere. Temperature inversions in the lower atmosphere (troposphere, 0-12km) can create ducts that trap VHF/UHF signals. This is most common over water or flat terrain, and is often accompanied by fog or haze conditions. It can extend VHF/UHF range to 1000+ km.

4.3.4 Earth-Moon-Earth (EME/Moonbounce)

It is possible to bounce signal off of Lunar. This requires high-gain antennas, high power, and very sensitive receivers. The path loss is very high (approx 250+ dB). This is common on 144 MHz, 432 MHz, and 1296 MHz.

4.3.5 Satellite Communication

Amateur radio satellites (OSCAR series, etc) provide a relay capability. This can be difficult due to Doppler shift, and the need to track the satellite and continuously adjust frequency.

5 Antennas and Feedlines

5.1 Antenna Basics

5.1.1 Introduction

An antenna converts electrical energy (RF current from a transmitter) into electromagnetic waves that travel through space. On receive, it does the reverse.The simplest (effective) antenna is a half-wave dipole. This is just a wire, half a wavelength long, split in the middle, and fed with coax.

5.1.2 The Half-Wave Dipole

The formula to calculate the length of a half-wave dipole is

$$L_{\text{metres}} = \frac{143}{f_{\text{MHz}}}.$$

This formula accounts for the end effect, where the electrical length is slightly shorter than the physical length.

> **Example Half-Wave Length.** These are some common half-wave lengths.
> For 40m band (7.1 MHz): 143/7.1 = 20.1 metres total.
> For 20m band (14.2 MHz): 143/14.2 = 10.1 metres total.
> For 2m band (146 MHz): 143/146 = 0.98 metres total.

5.1.3 Gain

Antenna gain doesn't create extra power, it focuses the existing power in a preferred direction. Because of conservation of energy, this results in less gain the non-focused direction.

Gain is measured in two ways

- dBi: Compared to an imaginary perfect point source (called an isotropic radiator)
- dBd: Compared to a half-wave dipole

$$dBi = dBd + 2.15$$

A dipole has 2.15 dBi (or 0 dBd).

5.1.4 Other Properties

Antennas have a few other properties.

Radiation resistance is the resistance that accounts for the power actually being radiated. For a half-wave dipole in free space, this is

$$R_{\text{rad}} \approx 73\Omega$$

This is close the 50Ω figure, but not exact. This is why a dipole has a slight SWR when fed with 50Ω coax. This discrepancy isn't enough to make it unusable.

Polarisation is another property. There are three types

- Horizontal: The dipole is mounted horizontally. This is standard for HF as the ionosphere scrambles polarisation anyway.
- Vertical: The dipole is mounted vertically. This is common for VHF/UHF mobile and base antennas.
- Circular: This is a dish, and is used on satellites as it handles random Faraday rotation in the ionosphere.

Cross-polarisation loss (horizontal antenna receiving a vertical signal) is about 20 dB, which is a significant loss. This is why matching polarisation matters.

The final property is effective radiated power (ERP). This is the power that appears to the be transmitter, taking antenna gain and feedline losses into account

$$ERP_{dBW} = P_{tx(dBW)} + G_{dBd} - L_{feedline(dB)}.$$

High-Power Yagi. With a 400W and a 3-element Yagi (7dBd) but a 1 dB feedline loss, the ERP is given by

$$26 + 7 - 1 = 32dBw = 1585W \text{ ERP}.$$

The 400W radio then looks like nearly 1600W in the antenna's main beam direction.

5.2 Directional Antennas

5.2.1 Introduction

A dipole radiates equally in two directions (broadside). If a station is in a specific direction, half the power is going the wrong way. This is why directional antennas are used, they focus the power (and receiving sensitivity) into a single direction.

5.2.2 Yagi-Uda Antenna

The most popular directional antenna for amateur radio is the Yagi-Uda (often called just Yagi) antenna. It has 3 main elements

- Reflector (behind): Slightly longer than $\lambda/2$, and reflects all the energy forward.
- Driven element (middle): This is about $\lambda/2$, and is the only element connected to the feedline.
- Directors (front): These are slightly shorter than $\lambda/2$, and are used to pull energy forward.

Table 6: Specs of Yagi Antennas

Elements	Gain (dBd)	Front-to-back	Notes
2	5	10 dB	Compact, good starter beam
3	7	15 dB	Most popular for 20m/15m/10m
5	9	20 dB	Good for series DX (distance)

Info. The front-to-back ratio is just as important as gain. It tells you how well the antenna rejects signals from behind. A 20 dB front-to-back ratio means stations behind are 100× weaker. This is good too reduce interference from any unwanted directions.

5.2.3 Quad Antenna

A quad antenna used full-wavelength square or diamond loops instead of straight elements. For the same boom length, a quad has significantly more gain than a Yagi and wider bandwidth. The drawback is that it is physically larger and harder to build.

5.2.4 Log-Period Dipole Array (LPDA)

This is the "broadband" directional antenna. Unlike a Yagi (which is designed for one band), and LPDA works across a wide range of frequencies, for example one antenna might be able to do 14-30 MHz. The downside of these is that the gain is lower than a Yagi on any single frequency. An LPDA gives maybe 5 dBd whereas a 3-element Yagi gives 7 dBd, but the LPDA covers many bands without switching.

5.3 Coax and Feedlines

5.3.1 Introduction

Feedlines carries RF from a transmitter to the antenna (and back for received signals). The most common type is a coaxial cable (coax), which is a center conductor surrounded by insulation, then a shield, then a jacket.

5.3.2 Characteristic Impedance

Every feedline has a characteristic impedance (Z_0) that is determined by its physical construction.

Table 7: Types of feedlines

Type	Z_0	Loss (per 30m @ 150 MHz)	Best for
RG-58	50Ω	6 dB (high)	Short runs, low power, patch leads
RG-213	50Ω	3 dB	General HF use, moderate runs
LMR-400	50Ω	1.5 dB (low)	VHF/UHF where loss matters
450Ω ladder line	0.3 dB (very low)	Balanced antennas, lowest loss.	

Info. A good rule of thumb for feedline is use the lowest-loss cable possible, especially for VHF/UHF where loss really matters. If 3 dB of a signal is lost in the coax, half of the total power is turned into heat.

5.3.3 Velocity Factor

RF travels slower in a feedline than in free space (by definition). The velocity factor tells you how much slower. Some typical values are

- Solid polyethylene coax: 0.66 (i.e. 66% speed of light)
- Foam dielectric coax: 0.8
- Air-spaced line: 0.95

If you need a quarter-wave matching section of coax, then you must account for the velocity factor. A $\lambda/4$ section at 14 MHz in solid coax is shorter than expected

$$\frac{300}{14/4} \times 0.66 = 3.54m,$$

not the expected 5.36 metres.

5.3.4 Standing Wave Ratio (SWR)

When the impedance isn't matched between the antenna and feedline, some power reflects back. This creates standing waves on the feedline, measured as SWR High SWR doesn't mean that all the reflected power is

Table 8: Affects of different SWRs

SWR	Reflected Power	Notes
1.0:1	0%	Perfectly matched, this is theoretical
1.5:1	4%	Excellent
2.0:1	11%	Good
3.0:1	25%	Needs attention (use a tuner)
$\infty : 1$	100%	Open or short circuit, nothing radiates

lost, it just bounces back and forth until its radiated or lost as heat in the feedline. The actual problem is feedline loss. With low-loss feedline, even a high SWR isn't necessarily terrible, but with a lossy one it is much worse.

5.4 Impedance Matching and Baluns

5.4.1 Introduction

A transmitter is designed to deliver power into 50Ω. If the antenna system presents something different, a matching network is needed to transform the impedance. Without one, the transmitter may reduce power (via SWR protection), or produce excessive reflected power.

5.4.2 Antenna Tuner (transmatch)

This is the most common matching device. It sits between a transmitter and feedline, and transforms whatever impedance the antenna presents back to the 50Ω the transmitter wants. There are three common topologies

- L-Network: Two components (series + parallel). This is the simplest, and works well for a moderate mismatch range.
- T-Network: This is three components. It has a wide matching range and is most popular in commercial tuners. Can handle very high SWR.
- Pi-Network: Again three components It can also provide some harmonic filtering, and is common in valve amplifier output stages.

> **Important.** An antenna tuner does not fix a bad antenna, it just tricks the transmitter into think everything is ok. The mismatch (and associated feedline losses) are still there between the tuner and antenna. The best approach is to make sure the antenna is resonant, then use the tuner for fine-tuning.

5.4.3 Quarter-Wave Transformer

A length of coax exactly $\lambda/4$ long with a specific impedance can transform one impedance to another. The require impedance of the matching section is then

$$Z_{\text{match}} = \sqrt{Z_1 \times Z_2}$$

> **Example.** Match 50Ω coax to a 200Ω folder dipole.
>
> $$\begin{aligned} Z &= \sqrt{50 \times^2 00} \\ &= \sqrt{10000} \\ &= 100\Omega. \end{aligned}$$
>
> This means you want to use a $\lambda/4$ section of 100Ω coax (or two pieces of 50Ω in series).

5.4.4 Baluns

Baluns is BALanced to UNbalacned. A dipole is balanced, i.e. the two halves are symmetrical. Coax is unbalanced because the shield is grounded. Connecting them directly would cause problems. RF current would flow on the outside of the coax shield, the feedline would radiate (distorting the antenna pattern), and RF would get into the shack/station, which would cause interference with other equipment.

This problem is solved using a balun. There are two common types

- 1:1 current balun: No impedance change, just forces balanced currents. This is best for dipoles fed with 50Ω coax.
- 4:1 balun: transforms 200Ω balanced to 50Ω unbalanced. This is good for folded dipoles and some loop antennas.

Tip. A simple and effective 1:1 current balun can be made by winding 10-12 turns of coax through a ferrite toroid at the antenna feedpoint. This chokes off common-mode current on the coax shield.

5.4.5 The Smith Chart

The Smith chart is a graphical tool for impedance matching problems. There are few things to know

- Centre = 50Ω, SWR = 1:1
- Circles around the centre = constant SWR
- Upper half = inductive (positive reactance)
- Lower half = capacitive (negative reactance)
- Moving clockwise = moving towards the transmitter along the feedline.

5.5 DIY Antennas

5.5.1 Introduction

One of the things allowed under the Advanced license is that you can design, build, and modify you own antennas. Home-built antennas can perform just as well as commercial ones, and will provide a greater understanding.

5.5.2 Half-Wave Dipole

This is the simplest effective HF antenna. All it needs is a wire, centre insulator, two end insulators, and some coax. To build it

1. Calculate the length: The total length (as given above) is given by $L = 143/f_{\text{MHz}}$ metres. Each leg of the antenna is half of that
2. Cut the wire: Use insulated or bare copper/steel wire. Cut a little longer than calculated
3. Centre insulator: this is where the coax is connected. Solder the centre conductor to one leg, the shield to the other. Add a 1:1 choke balun

6 EMC and Interference

6.1 Understanding and Fixing Interference

6.1.1 Introduction

With the Advanced license allowing for up to 400W, there two different sides that are likely to cause issues with EMI. The first is your signal interfering with others, such as TV interference (TVI), affecting neighbouring electronics, etc. The second is other thing interfering with you, such as plasma TVs, LED lights, switch mode power supplies, solar inverters, etc.

There are two main paths for interference to get into

- Conducted: interference travels along cables, such as power leads, audio cables, Ethernet cables, antenna feedlines. The cables act as unintentional antennas.
- Radiated: interference travels through air as electromagnetic waves directly into the affected device.

In amateur radio, conducted interference via common-mode currents is usually the culprit. RF gets onto the outside of cables and enters equipment where it's not wanted.

6.1.2 EMC Toolkit

Ferrite chokers are the most common fix. Snap-on or toroidal ferrite cores around cables are the most useful EMC tool. They block common-mode currents while having minimal effect on the wanted signal. They can be used on DC power leads to the radio, coax feedline at the entry to the shack, or cables on affected equipment.

Low-pass filters can be placed after the transmitter to suppress harmonics of the transmitted signal. A 7 MHz signal has harmonics at 14, 21, 28 MHz, etc. These must be suppressed to legal limits. A good LPF gives 40-60 dB suppression.

High-pass filters go on affected devices, such as on a neighbour's TV antenna input to block HF signal.

Band-pass filters can be used for specific problems. They pass only the operating frequency and reject everything else. This is the best filtering, but only works on one band.

6.1.3 Intermodulation

If you overdrive your amplifier, it creates intermodulation product. These are spurious signals that interfere with other users. To prevent this, keep drive levels within the linear range of the PA.

6.2 EMR Safety

6.2.1 Introduction

At 400W with a directional antenna, you are generating significant RF fields near the antenna. Radio waves at sufficient intensity can cause tissue heating, and legally your station has to comply with safety limits.

In the far field, power density follows the inverse square law

$$S = \frac{P \times G}{4\pi d^2} \mathrm{W/m^2}$$

Where P is power (W), G is the antenna gain (linear, not dB), and d is the distance (m).

Example. 400W into a Yagi with 10 dBi gain (=10 linear):

- At 5 metres: $S = (400 \times 10)/(4\pi \times 25) = 12.7\mathrm{W/m^2}$. This potentially exceeds limits.
- At 10 metres: $S = (400 \times 10)/(4\pi \times 100) = 3.2\mathrm{W/m^2}$
- At 20 metres: $S = (400 \times 10)/(4\pi \times 400) = 0.8\mathrm{W/m^2}$. This is safe for most limits

We can rearrange this for distance to get

$$d = \sqrt{\frac{P \times G}{4\pi \times S_{\mathrm{limit}}}}.$$

Rule of thumb for 400W. Keep people at least 10-15 metres from a directional HF antenna during transmission. The exact distance depends on the gain of the antenna and the applicable exposure limit for the frequency.

6.2.2 Australian Requirements

You must comply with ARPANSA RPS3 (Radiation Protection Standard). Key points from this

- Exposure limits vary with frequency
- General public limits are stricter than occupational limits
- Duty cycle matters. SSB and CW have lower average power than FM because they aren't a continuous transmission.

There are some practical safety measures

- Keep antennas high. This reduces exposure at ground level
- Use minimum power
- Consider duty cycle
- Restrict access to areas near antennas during transmission
- Never touch antenna elements or feedlines during transmission.

> **Near-field wanring.** The above formula are for far-field. Close the antenna, fields are complex and can be much stronger than the formula predicts. On 80m, a few wavelengths is 500+ metres, so for HF antennas you may always be in the near field at ground level. Use conservative estimates.

6.2.3 Worked EMC Examples

These are similar to questions on the exam.

> **Is my antenna safe. Problem**: You're running 400W PEP into a 3-element Yagi with 7 dBi gain. Your neighbour's fence is 8 metres from the antenna. The exposure limit for your frequency is 2W/m^2. Is it safe?
>
> **Answer**
>
> 1. Convert gain from dBi to linear: $7\text{dBi} = 10^(7/10) = 5.01$
> 2. Calculate the power density: $S = PG/(4\pi d^2) = (400 \times 5.01)/(4\pi \times 64) = 2.94\text{W/m}^2$
> 3. Compare to limit: $2.49 > 2.0\text{W/m}^2$
>
> This is not safe to operate like this.

> **What's the safe distance. Problem**: Same setup, but now find the minimum safe difference for this limit.
>
> **Answer**
>
> 1. Use the distance formula: $d = \sqrt{PG/(4\pi S_{\text{limit}})}$
> 2. Substituting the values: $d = \sqrt{(400 \times 5.01)/(4\pi 2)} = 8.93$ metres.

> **How much harmonic suppression is needed. Problem**: Your 7 MHz transmitter produces a harmonic at 14 MHz that's 30 dB below the fundamental. The legal limit requires harmonics to be at least 50 dB below. How much additional filtering is needed.
>
> **Answer**: You need an additional $50 - 30 = 20$ dB of suppression. A simple 3-element low-pass filter (3 poles times -20 dB/decade) would be marginal at one octave above cutoff. A 5-element filter would give a comfortable margin.

7 Measurements and Test Equipment

7.1 RF Power and Antenna Measurements

7.1.1 Measuring RF Power

There are two common pieces of equipment used for measuring RF power.

The first is the directional wattmeter. It measures forward power (power going to the antenna) and reflected power simultaneously. The net power to the antenna is then forward power minus reflected power. This also allows SWR to be calculated.

The other is a dummy load. This is a non-radiating 50Ω resistive load for testing transmitters without going on air. This allows for checking transmitter output power, adjusting drive levels, and testing without causing interference.

7.1.2 Power in Decibels

Instead of saying 0.001 watts or 400 watts, you can also use decibels referenced to a standard power level.

Table 9: Different ways to express power in decibels

dBm (ref: 1 mW)	dBW (ref: 1 W)	Actual Power
0 dBm	-30 dBW	1 mW
+30 dBm	0 dBW	1 W
+40 dBm	+10 dBW	10 W
+50 dBm	+20 dBW	100 W
+56 dBm	+26 dBW	400 W

7.1.3 Antenna Analyser

An antenna analyser sweeps across a frequency range and shows

- SWR vs frequency: shows where the antenna is resonant and how wide the usable bandwidth is
- Impedance: tells you if the antenna needs to be longer (capacitive) or short (inductive)
- Return loss: another way to express how well the antenna matches 50Ω

7.1.4 Frequency Counter

A frequency counter directly measures the frequency of a signal. The gate time determines the resolution. A 1 second gate means 1 Hz resolution, a 10 second gate means 0.1 Hz resolution. GPS-disciplined counters provide the highest accuracy timebase.

7.2 Practical Measurement Scenarios

My SWR is suddenly high on 20m. Tools needed: antenna analyser.

1. Disconnect the coax from the radio and connect the antenna analyser
2. Sweep 13-15 MHz and look at the SWR curve
3. If SWR is high everywhere: possible broken feedline or open connection, check connectors and cable
4. If SWR dip has shifted frequency: something has changed the antenna (i.e. damage). The analyser shows you where the resonance has moved to
5. If the impedance shows high R (e.g. 150Ω): possible feedline issue, water in the coax can change the impedance

Someone reports my signal has splatter. Tools needed: Spectrum analyser or SDR waterfall, dummy load

1. Connect transmitter to dummy load through a directional coupler or use a loose coupling to the

spectrum analyser

2. Transmit into the dummy load while speaking normally
3. Watch the spectrum. The signal should be clean with defined edges
4. If you see broad shoulders spreading beyond signal bandwidth, reduce mic gain or drive level
5. Check that ALC is not being driven excessively

I want to check my transmitter's harmonics. Tools needed: spectrum analyser, dummy load, attenuator

1. Connect transmitter to the dummy load through a low-pass filter, with a tap or coupler to the spectrum analyser
2. Use an attenuator between the coupler and spectrum analyser to avoid overloading it
3. Transmit a carrier and look for signals at 2, 3, and 4 times your fundamental frequency
4. Harmonics should be at least 40-50 dB below the fundamental
5. If they're not, the low-pass filter may be faulty or the wrong cutoff frequency

Table 10: Common power in measurement questions

Power	dBm	dBW
1 mW	0 dBm	-30 dBW
10 mW	+10 dBm	-20 dBW
100 mW	+20 dBm	-10 dBW
1 W	+30 dBm	0 dBW
10 W	+40 dBm	+10 dBW
100 W	50 dBm	+20 dBW
400 W	+56 dBm	+26 dBW